

Sustaining Collective Welfare Under Disruption: A Pilot Study of Cooperative Resilience in Human and LLM Multi-Agent Systems

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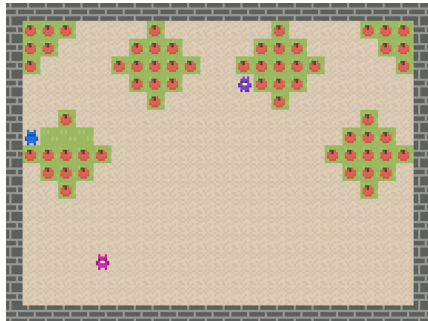
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Motivation

Individual

→
Goal: Maximize the number of apples collected



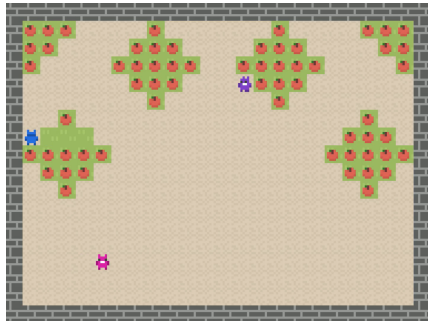
Collective

←
Goal: Avoid depleting shared resources

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Collective

←
Goal: Avoid depleting shared resources

Let's imagine a **disruption**. If we still care about preserving cooperation and collective well-being during these events, then we are talking about **cooperative resilience**.

Outline

- 1 What Should We Preserve?
- 2 How Can Humans and LLM Agents Be Compared?
- 3 Who Is More Resilient?
- 4 What Do We Learn?

What Should We Preserve?

Research Question

How do human groups and LLM-based multi-agent systems sustain collective welfare under disruption?

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Humans

Human participants
interacting in a shared-resource dilemma



Communication

VS.

LLM Agents

LLM-based agents
interacting in the same social dilemma



Communication

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Communication

What role does communication play in cooperative resilience?

What do we mean by Cooperative Resilience?

Cooperative Resilience

Cooperative resilience is the ability of a system, involving the collective action of individuals—whether humans, machines, or both—to anticipate, prepare for, resist, recover from, and transform in the face of disruptive events that threaten their collective welfare.

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**The objective is not only to resist disruption,
but to preserve collective welfare throughout it.**

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What does Collective Welfare mean in this Social Dilemma?

In the Commons Harvest dilemma, collective welfare requires **preserving the resource while maintaining fair and sustained access.**

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Are enough resources
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Long-term
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Is the regenerative
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Fairness in
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Are resources shared
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Satiation

Sustained
resource access

Can agents keep
consuming over time?

What Disrupts Collective Welfare?

We expose the system to **two sources of pressure**.

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Persistent Pressure



Unsustainable RL Agent

A fixed agent continuously harvests resources without following a sustainable strategy.

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Environmental Shocks



Resource Removal Events

Apples are intermittently removed from the environment, reducing the available shared resource.

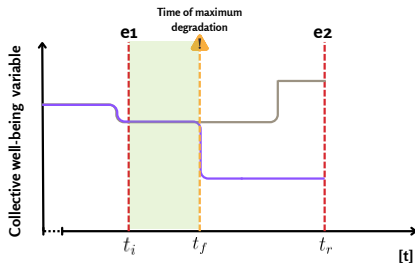
How do we measure Cooperative Resilience?

For each collective-welfare indicator, we compare the disrupted trajectory against a shared reference.

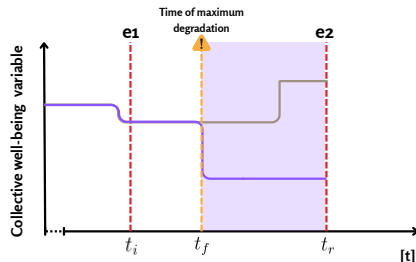
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Failure Profile



Recovery Profile



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Failure Profile

$$FP_k = \frac{\int_{t_d}^{t_f} I_k^{\text{disrupted}}(t) dt}{\int_{t_d}^{t_f} I_k^{\text{baseline}}(t) dt}$$

How much performance is retained
from the disruption to maximum degradation.

Recovery Profile

$$RP_k = \frac{\int_{t_f}^{t_r} I_k^{\text{disrupted}}(t) dt}{\int_{t_f}^{t_r} I_k^{\text{baseline}}(t) dt}$$

How much performance is sustained/recovered
after maximum degradation.

Resilience Score

1. Resilience for each welfare indicator

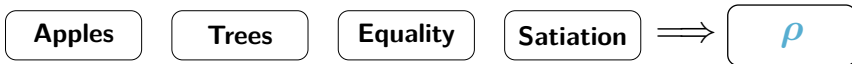


$$\rho_k = \frac{t_d + FP_k \Delta t_f + RP_k \Delta t_r}{t_d + \Delta t_f + \Delta t_r}$$

Failure and recovery are weighted by the **duration of each phase**.

Resilience Score

2. Combine all welfare dimensions

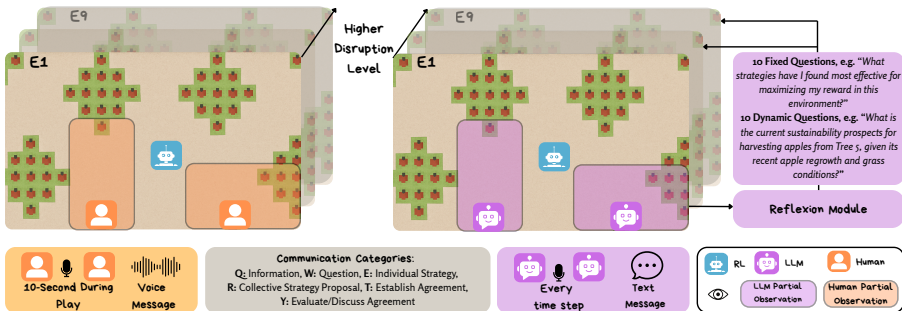


$$\rho = \left(\frac{1}{K} \sum_{k=1}^K \frac{1}{\rho_k} \right)^{-1}$$

The harmonic mean penalizes poor resilience in any welfare dimension.

How Can Humans and LLM Agents Be Compared?

Experimental Setup



Four conditions

- Humans / Humans + Communication
- LLM Agents / LLM Agents + Communication

Shared across conditions

- Same social dilemma and action space
- Same disruptions and welfare indicators

Disruption Curriculum

Increasing disruption severity



$$d \in \{1, 2, 3\}$$

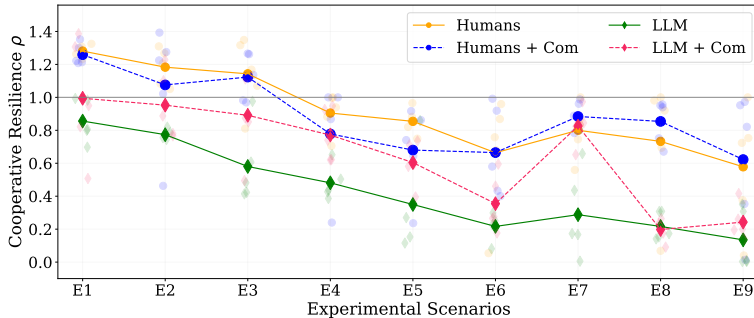
$$v_s \in \{0.3, 0.5, 0.7\}$$

d : number of disruptive events

v_s : probability of removing individual apples

Who Is More Resilient?

Humans sustain higher Cooperative Resilience under disruption



Humans

High and relatively stable resilience

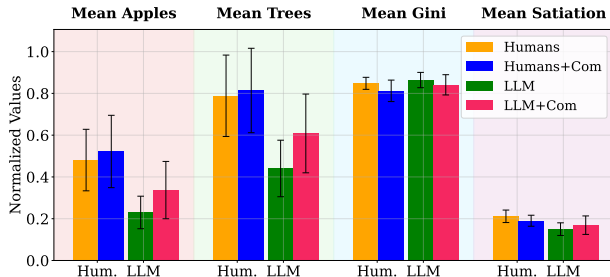
LLM Agents

Resilience decreases with disruption severity

Communication

Largest benefit for LLM-based agents

What explains the Resilience Gap?



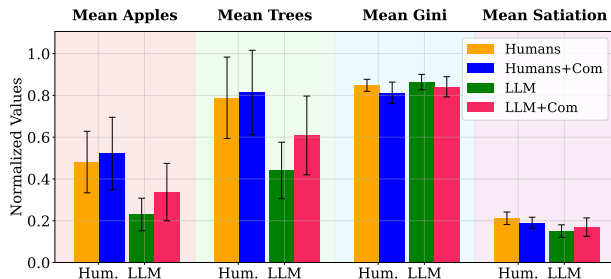
Humans

- Higher resource availability
- More trees preserved
- Higher collective satiation

LLM Agents

- Lower resource availability
- Fewer trees preserved
- Lower collective satiation

What explains the Resilience Gap?



Humans

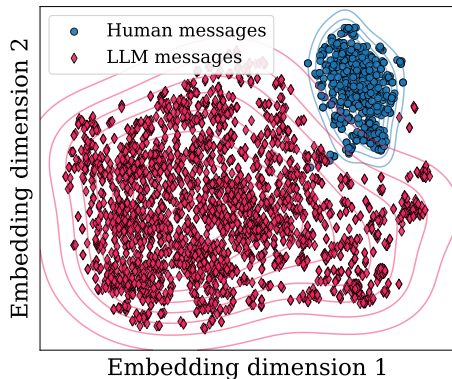
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Higher resilience is closely associated with more sustainable resource management.

Semantic Organization of Communication



Human messages

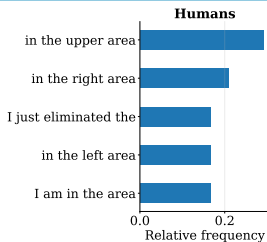
- More concentrated projected distribution
- Stronger local structure

LLM messages

- More dispersed projected distribution
- Consistent with greater linguistic variability

Messages were embedded using a **multilingual encoder** and projected with t-SNE.

From Language to Coordination

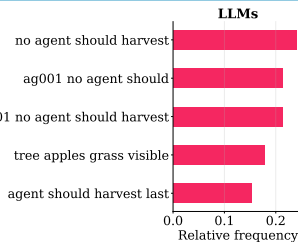


Humans

Spatial and action-oriented communication

"upper area"

"right area"



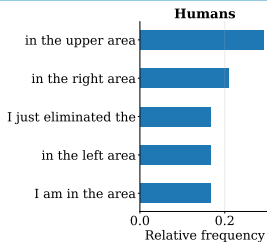
LLM Agents

More abstract and normative communication

"no agent should harvest"

"agent should harvest last"

From Language to Coordination

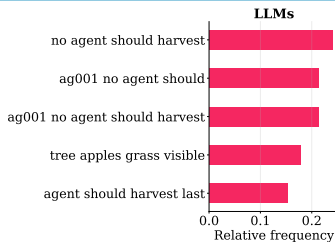


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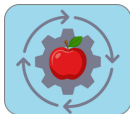
"agent should harvest last"

Communication content does not necessarily translate into grounded collective action.

What Do We Learn?

Interpreting the Performance Difference

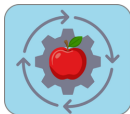
Interpreting the Performance Difference



Sustainable Resource Management

Preserve resources
and regenerative capacity

Interpreting the Performance Difference

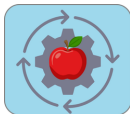


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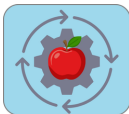


Behavioral Adaptation

Adjust behavior as
disruptions intensify

+

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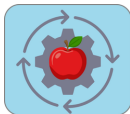
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Grounded Coordination

Connect communication
with local actions

Interpreting the Performance Difference



Sustainable Resource Management

Preserve resources
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Behavioral Adaptation

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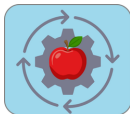


Grounded Coordination

Connect communication
with local actions

Higher Cooperative Resilience

Interpreting the Performance Difference



Sustainable Resource Management

Preserve resources
and regenerative capacity

+



Behavioral Adaptation

Adjust behavior as
disruptions intensify

+



Grounded Coordination

Connect communication
with local actions

Higher Cooperative Resilience

High-level reasoning and communication do not necessarily
translate into effective collective action.

Limitations and Scope

This study provides exploratory, system-specific evidence.

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Small Sample

Five trials
per condition

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Commons Harvest
social dilemma

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Specific Agents

One LLM architecture
and prompting setup

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Different Interfaces

Population-specific
observation and
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The observed human advantage should not be interpreted as a general causal difference between humans and LLMs.

What should we take from this?

1. **Cooperation under nominal conditions is not enough.** Multi-agent systems should also be evaluated by how they preserve collective welfare under disruption.
2. **Human groups showed higher cooperative resilience in this setting.** Their behavior was associated with more sustainable resource use, adaptation, and grounded coordination.
3. **Communication helped LLM-based agents, but did not close the gap.** Communication alone does not guarantee effective collective action.

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Cooperative AI should remain cooperative when things go wrong.

Thank you

Thank you for your attention!

Questions?

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*Sustaining Collective Welfare Under Disruption:
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Backup: Statistical Analysis

Exploratory non-parametric comparisons across disruption scenarios

Comparison	Cliff's δ	Uncorrected sig.
Humans vs. LLMs	0.85	6/9
Humans vs. Humans + Com	0.10	0/9
LLMs vs. LLMs + Com	-0.54	2/9
Humans + Com vs. LLMs + Com	0.48	1/9

- Mann–Whitney U tests were used across scenarios.
- No comparison remained significant after Bonferroni correction.
- Effect sizes therefore provide the most informative exploratory signal.

Large human–LLM differences, but exploratory evidence only.

Backup: Shared Reference

Why not use population-specific baselines?

If each population used its own baseline, resilience scores would be **relative to different reference behaviors**.

Therefore, values would **not be directly comparable** across humans and LLM-based agents.

Goal: a common evaluation scale

1. Group baseline trajectories

2. Mean trajectory
across five episodes

3. Temporal median
for each group

4. Median across
group-level references
Shared baseline for all conditions

Backup: Human and LLM Interaction Protocols

Humans

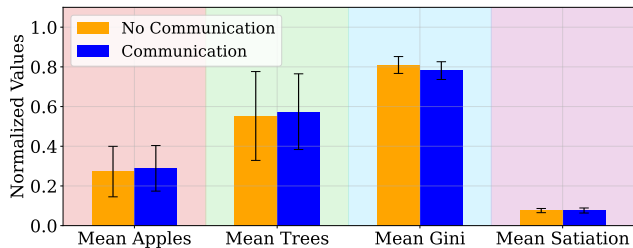
- Visual partial observation
- Keyboard interaction
- Spanish voice communication
- Communication in 10-second intervals

LLM Agents

- Textualized partial observation
- Text-conditioned action selection
- English text communication
- Communication at each time step
- Reflection + persistent memory

**The conditions share the task and disruption protocol,
but are not modality- or architecture-matched.**

Backup: Long-Horizon Human Evaluation



More demanding setting

- Horizon: 1000 steps
- Reduced apple regrowth
- Sequential E1–E9 curriculum

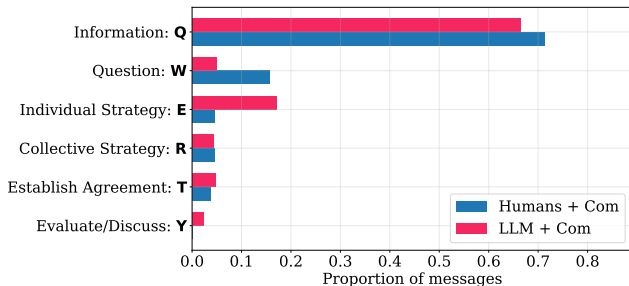
LLM agents

- Exhaust resources around 250 steps

Human groups

- Sustain resources over the full horizon

Backup: Communication Categories



- Both populations predominantly exchanged **information**.
- Humans asked relatively more questions.
- LLM agents communicated relatively more about their own intended actions.
- Explicit agreement-oriented categories remained uncommon in both groups.